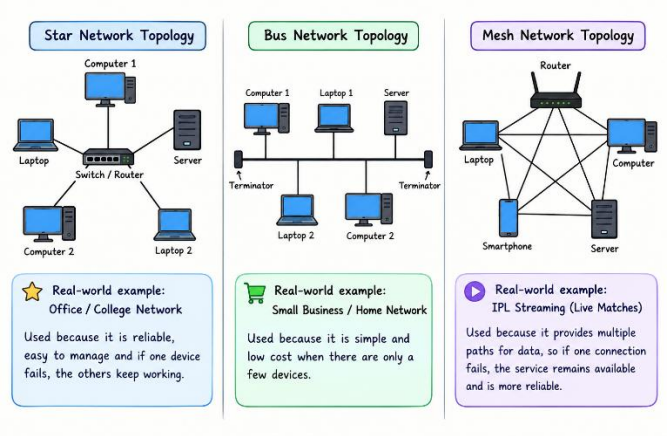
NETWORK TOPOLOGISE

1.Draw and label diagrams of star, bus, and mesh network topologies on paper or using any drawing tool. For each, write one real-world app or service (like WhatsApp, Flipkart, or IPL streaming) that could use that topology and explain why in 1-2 lines.

ANS:



1. Star Topology

Real-world example: College computer lab network.

Explanation: Star topology is easy to manage and troubleshoot. If one device's connection fails, the other devices can continue working.

2. Bus Topology

Real-world example: A small network using older Ethernet bus cabling.

Explanation: Bus topology is simple and uses less cable. However, a fault in the main cable can disrupt the whole network.

3. Mesh Topology

Real-world example: Mesh Wi-Fi network used for live IPL streaming.

Explanation: Mesh topology provides multiple paths for data. If one connection fails, data may travel through another path, improving reliability.

2.Create a comparison table listing at least four pros and four cons for each of these topologies: star, bus, ring, mesh. Include a column for 'Best Use Case' and give an example from apps you use (e.g., Instagram, Zomato, Paytm).

ANS:

Comparison of Network Topologies

Topology	Pros (Advantages)	Cons (Disadvantages)	Best Use Case with Example

Star	1. Easy to install. 2. Easy to troubleshoot. 3. Failure of one device doesn't affect others. 4. Easy to add new devices.	1. Requires more cables. 2. Higher installation cost. 3. Central switch failure stops network communication. 4. Performance depends on the central device.	College computer lab: Computers connect through a central switch to access the Internet and services such as Instagram.
Bus	1. Uses less cable. 2. Low installation cost. 3. Simple for small networks. 4. Doesn't require a central switch.	1. Main cable failure can stop the network. 2. Troubleshooting is difficult. 3. Performance decreases with heavy traffic. 4. Limited cable length and number of devices.	Small legacy LAN: Older Ethernet networks used bus cabling to connect computers accessing websites such as Zomato.
Ring	1. Data travels in an organized direction. 2. Predictable access to the network. 3. Can reduce data collisions in token-based rings. 4. Each device gets a turn to transmit in token-ring systems.	1. A single link or device failure can disrupt a basic ring. 2. Adding or removing devices may interrupt service. 3. Troubleshooting can be difficult. 4. Delay can increase as the number of devices grows.	Ring-based campus network: A ring can connect campus network switches for services such as Paytm access.
Mesh	1. Highly reliable. 2. Provides multiple data paths. 3. Can reroute traffic after a link failure. 4. Supports high availability.	1. Expensive to install. 2. Requires more cables and ports. 3. Complex to configure and maintain. 4. Uses more network equipment and resources.	Internet backbone or mesh Wi-Fi: Multiple paths help keep Instagram or IPL streaming services accessible when a connection fails.

3.Imagine you are designing a new multiplayer mobile game like PUBG or Free Fire. Choose a suitable network topology for the game servers and justify your choice in 3-4 lines, mentioning how it would impact speed, reliability, and cost.

ANS:

Suitable Network Topology for a Multiplayer Mobile Game

Chosen Topology: Mesh Topology

1. Mesh topology provides multiple communication paths, helping players enjoy fast and smooth gameplay with low latency.
2. It improves reliability because traffic can use an alternative path if one network connection fails.
3. It can support continuous multiplayer gaming with fewer interruptions and better connectivity.
4. However, mesh topology is expensive to install and maintain because it requires more network links and equipment.

**4.Given this scenario: A small Flipkart-style warehouse wants to connect 8 computers for order processing and inventory updates. Recommend a network topology, sketch a quick diagram, and explain your reasoning in 2-3 lines.

Hint: Consider factors like cost, ease of troubleshooting, and future expansion.**

ANS:

Recommended Network Topology: Star Topology

Diagram: Connecting 8 Computers in a Warehouse

Each computer connects to one central network switch using an Ethernet cable.